

Car Performance & Tuning Lab

System Analysis & Design:

1. Problem Statement & Objectives – Define the problem being solved and project goals:

1) Problem Statement:

Car enthusiasts don't have a reliable way to predict how modifications (like turbo, exhaust, ECU remap) will affect car performance before actually applying them. This leads to uncertainty, wasted money, and trial-and-error decisions.

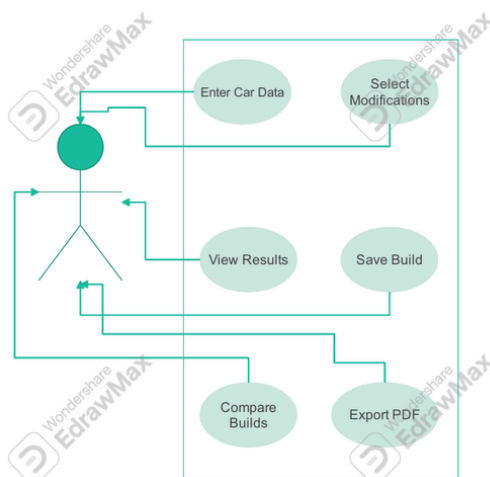
#Objectives:

- Build a system that simulates car performance after modifications
- Provide real-time calculations for:
 - 0–100 km/h time
 - Top speed
 - Fuel consumption
- Allow users to compare stock vs modified cars
- Help users make better decisions before applying changes

2) Use Case Diagram:

<u>Actor:</u> User	<u>Use Cases:</u> ● Enter car specifications (weight, horsepower, torque) ● Select drivetrain (FWD / RWD / AWD) ● Choose modifications (Turbo, Exhaust, ECU Remap, etc.) ● View performance results ● Compare builds ● Save configurations ● Export results as PDF
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USECASE DIAGRAM



3)Functional & Non-Functional Requirements:

Functional Requirements:

- User inputs car data
- System calculates performance metrics
- System updates results in real-time
- User can select multiple modifications
- System displays charts and comparisons
- User can save and export builds

Non-Functional Requirements:

- Fast response (real-time updates)
- High accuracy in calculations
- Responsive UI (works on mobile & desktop)
- Multi-language support (Arabic & English)
- Scalability for future features

4) Software Architecture:

Architecture Style:

- Client-Server (MVC-like)

Components:

- Frontend: React + TypeScript
- Backend: Node.js + Express
- Database: (for saved builds – e.g. MongoDB or SQL)

Interaction:

- User → Frontend → API → Backend → Response → UI update

2. Database Design & Data Modeling:

ER Diagram:

Entities:

- User
- Car
- Modification
- Build

Relationships:

- User creates many Builds
- Build contains one Car
- Build has many Modifications

Logical Schema

<u>User:</u>	<u>Car:</u>	<u>Build:</u>	<u>Modification:</u>
• user_id (PK) • name • email	• car_id (PK) • weight • horsepower	• build_id (PK) • date / time • user_id (FK) • car_id (FK)	• mod_id (PK) • type • cost

The Diagram:

ER Diagram- Car System

